

2. | 기계항공공학 3학년

3.  $y'' - y = 0$  20200809 이종현

let  $y = e^{\lambda x}$

$$(\lambda^2 - 1)e^{\lambda x} = 0$$

$$\lambda = \pm 1$$

general sol.  $y = C_1 e^{-x} + C_2 e^x$

b.  $xy'' + 2y' + xy = 0 \rightarrow y = ?$

$$y'' + \frac{2}{x}y' + y = 0$$

$$y_1 = \frac{\cos x}{x}$$

$$u = \int \frac{C}{y_1^2} e^{-\int P dx}$$

$$= \int \frac{C x^2}{\cos^2 x} e^{-\int \frac{2}{x} dx} dx$$

$$= \int \frac{C x^2}{\cos^2 x} e^{-2 \ln |x|} dx$$

$$= \int \frac{C}{\cos^2 x} dx$$

$$= C \tan x$$

$$\therefore y_2 = u y_1$$

$$= C \tan x \times \frac{\cos x}{x}$$

$$y_2 = C \frac{\sin x}{x}$$

$$\therefore \text{general solution } y = C_1 \frac{\cos x}{x} + C_2 \frac{\sin x}{x}$$

1b.  $y'' + 2y' + y = 0, y(0) = 2, y'(0) = -1$

a)  $e^x, xe^x$

$$w(e^x, xe^x) = \begin{vmatrix} e^x & xe^x \\ -e^x & e^x - xe^x \end{vmatrix}$$

$$= e^x(e^x - xe^x) - xe^x(-e^x)$$

$$\neq 0$$

$$\therefore \text{linearly independent}$$

let  $y = C_1 e^x + C_2 x e^x$

$$y' = C_1 e^x + C_2 e^x + C_2 x e^x$$

$$y'' = C_1 e^x + C_2 e^x + C_2 x e^x$$

$\rightarrow$  plug in

$$C_1 e^x + C_2 x e^x + 2(C_1 e^x + C_2 x e^x) + C_1 e^x + C_2 x e^x = 0$$

$$\underbrace{(C_1 + 2C_1 + C_1)}_0 e^x + \underbrace{(C_2 + 2C_2 + C_2)}_0 x e^x = 0$$

$\therefore$  given functions form a basis of solutions

b)  $y'' + 2y' + y = 0, y(0) = 2, y'(0) = -1$

let  $y = e^{\lambda x}$

$$(\lambda^2 + 2\lambda + 1)e^{\lambda x} = 0$$

$$\lambda = -1$$

$$\therefore y_1 = e^{-x}$$

Reduction of order

$$y_2 = u y_1$$

$$u = \int \frac{C}{y_1^2} e^{-\int P dx} dx$$

$$= \int C e^{2x} e^{-2x} dx$$

$$= Cx$$

$$\therefore y_2 = x e^{-x}$$

$\therefore$  general solution

$$y = C_1 e^{-x} + C_2 x e^{-x}$$

$$y(0) = 2$$

$$\therefore 2 = C_1$$

$$y' = C_1 e^{-x} + C_2 e^{-x} - C_2 x e^{-x}$$

$$y'(0) = -1$$

$$-1 = -C_1 + C_2$$

$$1 = C_2$$

$$\therefore y = 2e^{-x} + x e^{-x}$$

2.2

2.  $y'' + 36y = 0$

let  $y = e^{\lambda x}$

$(\lambda^2 + 36)e^{\lambda x} = 0$

$\lambda^2 = -36$

$\lambda = \pm 6i$

$y = e^{-6ix}, e^{6ix}$

$\therefore$  general solution  $y = C_1 e^{-6ix} + C_2 e^{6ix}$   
 $= C_1 (\cos 6x - i \sin 6x) + C_2 (\cos 6x + i \sin 6x)$   
 $= A \cos 6x + B \sin 6x$

8.  $y'' + y' + 3.25y = 0$

let  $y = e^{\lambda x}$

$(\lambda^2 + \lambda + 3.25)e^{\lambda x} = 0$

$\lambda = \frac{-1 \pm \sqrt{1-13}}{2}$

$y = C_1 e^{-\frac{1}{2}x} \cdot e^{\sqrt{3}ix} + C_2 e^{-\frac{1}{2}x} \cdot e^{-\sqrt{3}ix}$   
 $= e^{-\frac{1}{2}x} (C_1 e^{\sqrt{3}ix} + C_2 e^{-\sqrt{3}ix})$   
 $= e^{-\frac{1}{2}x} (A \cos \sqrt{3}x + B \sin \sqrt{3}x)$

12.  $y'' + 8y' + 15y = 0$

let  $y = e^{\lambda x}$

$(\lambda^2 + 8\lambda + 15)e^{\lambda x} = 0$

$\lambda_{1,2} = -3, -5$

$\therefore y = C_1 e^{-3x} + C_2 e^{-5x}$

13.  $9y'' - 30y' + 25y = 0$

let  $y = e^{\lambda x}$

$(9\lambda^2 - 30\lambda + 25)e^{\lambda x} = 0$

$(3\lambda - 5)^2 = 0$

$\lambda = \frac{5}{3}$

$\therefore y = C_1 e^{\frac{5}{3}x} + C_2 x e^{\frac{5}{3}x}$

15.  $y'' + 0.54y' + (0.0729 + a)y = 0$

let  $y = e^{\lambda x}$

$\{ \lambda^2 + 0.54\lambda + (0.0729 + a) \} e^{\lambda x} = 0$

$\lambda = -0.27 \pm 1.17i$

$\therefore y = C_1 e^{-0.27ix} e^{1.17ix} + C_2 e^{-0.27ix} e^{-1.17ix}$   
 $= e^{-0.27ix} (C_1 e^{1.17ix} + C_2 e^{-1.17ix})$   
 $= e^{-0.27ix} (A \cos 1.17ix + B \sin 1.17ix)$

16.  $y_1 = e^{2.6x} \quad y_2 = e^{-4.3x}$

$y_1, y_2$  are solutions

$(2.6)^2 e^{2.6x} + 2.6a e^{2.6x} + b e^{2.6x} = 0$   
 $\{ (2.6)^2 + 2.6a + b \} e^{2.6x} = 0$   
 $2.6^2 + 2.6a + b = 0$

$y_2$  is also a solution

$(-4.3)^2 e^{-4.3x} - 4.3a e^{-4.3x} + b e^{-4.3x} = 0$   
 $\{ (4.3)^2 - 4.3a + b \} e^{-4.3x} = 0$   
 $(4.3)^2 - 4.3a + b = 0$

subtracting

$a = \frac{19}{10}, b = -\frac{559}{50}$   
 $\therefore y'' + \frac{19}{10}y' - \frac{559}{50}y = 0$

21.  $y'' + 9y = 0, y(0) = 0.2, y'(0) = -1.5$

let  $y = e^{\lambda x}$

$(\lambda^2 + 9)e^{\lambda x} = 0$

$\lambda = \pm 3i$

$\therefore y = C_1 e^{3ix} + C_2 e^{-3ix}$

$= A \cos 3x + B \sin 3x \quad \leftarrow y(0) = 0.2$

$0.2 = A$

$y' = -3A \sin 3x + 3B \cos 3x \quad \leftarrow y'(0) = -1.5$

$-1.5 = 3B$

$-0.5 = B$

$\therefore y = 0.2 \cos 3x - 0.5 \sin 3x$

23.  $y'' - 3y' - 4y = 0, y(0) = 2, y'(0) = 1$

let  $y = e^{\lambda x}$

$(\lambda^2 - 3\lambda - 4)e^{\lambda x} = 0$

$\lambda_{1,2} = 4, -1$

$\therefore y = C_1 e^{4x} + C_2 e^{-x} \quad \leftarrow y(0) = 2$

$2 = C_1 + C_2$

다들

$$y = c_1 e^{4x} + c_2 e^{-x}$$

$$y' = 4c_1 e^{4x} - c_2 e^{-x} \leftarrow y'(0) = 1$$

$$1 = 4c_1 - c_2$$

gegeben

$$c_1 = \frac{3}{5}, c_2 = \frac{7}{5}$$

$$\therefore y = \frac{3}{5} e^{4x} + \frac{7}{5} e^{-x}$$

2.4

```
function xdot = sdof(t,x);

m = 1; k = 5; c = 6;

xdot = zeros(2,1);

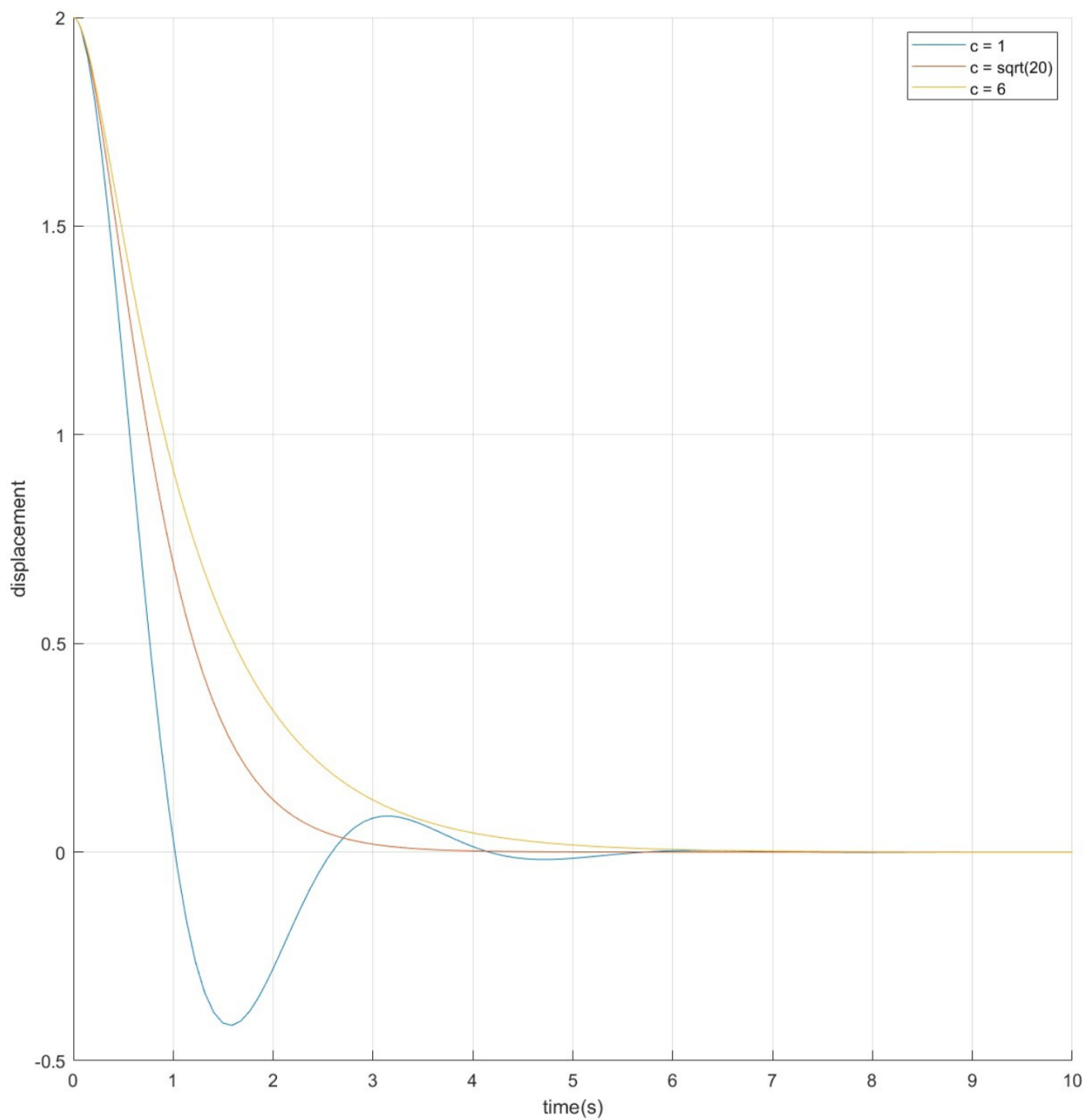
xdot(1) = x(2);

xdot(2) = -(k/m)*x(1)-(c/m)*x(2);

end
```

```
[t,x] = ode45(@sdof,[0 10],[2 0])
```

를 작성하여  $c$ 를 바꾸어 data를 얻었다





2.5

$$5. 4x^2y'' + 5y = 0$$

$$\text{let } y = x^m$$

$$\{4m(m+1) + 5\} x^m = 0$$

$$4m^2 - 4m + 5 = 0$$

$$m = \frac{2 \pm \sqrt{4 - 20}}{4} = \frac{1}{2} \pm i$$

$$\therefore y = c_1 x^{\frac{1}{2} + i} + c_2 x^{\frac{1}{2} - i}$$

$$= x^{\frac{1}{2}} (c_1 x^i + c_2 x^{-i})$$

$$= x^{\frac{1}{2}} (c_1 e^{i \ln x} + c_2 e^{-i \ln x})$$

$$= x^{\frac{1}{2}} (A \cos(\ln x) + B \sin(\ln x))$$

$$6. x^2 y'' + 0.7 x y' - 0.1 y = 0$$

$$\text{let } y = x^m$$

$$x^m \{m(m+1) + 0.7m - 0.1\} = 0$$

$$m^2 - 0.3m - 0.1 = 0$$

$$10m^2 - 3m - 1 = 0$$

$$m_{1,2} = \frac{1}{2}, -\frac{1}{5}$$

$$\therefore y = c_1 x^{\frac{1}{2}} + c_2 x^{-\frac{1}{5}}$$

$$8. (x^2 D^2 - 3x D + 3I) y = 0$$

$$x^2 y'' - 3x y' + 3y = 0$$

$$\text{let } y = x^m$$

$$\{m(m-1) - 3m + 3\} x^m = 0$$

$$m^2 - 4m + 3 = 0$$

$$m_{1,2} = 1, 3$$

$$\therefore y = c_1 x^1 + c_2 x^3$$

$$12. x^2 y'' - 4x y' + 6y = 0, y(1) = 0.4, y'(1) = 0$$

$$\text{let } y = x^m$$

$$\{m(m-1) - 4m + 6\} x^m = 0$$

$$m^2 - 5m + 6 = 0$$

$$m_{1,2} = 2, 3$$

$$\therefore y = c_1 x^2 + c_2 x^3 \quad \leftarrow y(1) = 0.4$$

$$0.4 = c_1 + c_2$$

$$y' = 2c_1 x + 3c_2 x^2 \quad \leftarrow y'(1) = 0$$

$$0 = 2c_1 + 3c_2$$

$$\therefore c_1 = -0.8$$

$$c_2 = 1.2$$

$$\therefore y = -0.8x^2 + 1.2x^3$$

$$14. x^2 y'' + x y' + 4y = 0, y(1) = 0, y'(1) = 2.5$$

$$\text{let } y = x^m$$

$$\{m(m+1) + m + 4\} x^m = 0$$

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

$$\therefore y = c_1 x^{2i} + c_2 x^{-2i}$$

$$= c_1 e^{2i \ln x} + c_2 e^{-2i \ln x}$$

$$= A \cos(2 \ln x) + B \sin(2 \ln x) \quad \leftarrow y(1) = 0$$

$$0 = A$$

$$y' = 2B \sin(2 \ln x) \times \frac{1}{x}$$

$$2.5 = 2B$$

$$\frac{5}{2} = B$$

$$\therefore y = \frac{5}{2} \sin(2 \ln x)$$

$$15. x^2 y'' - x y' + y = 0, y(1) = 1.5, y'(1) = 0.25$$

$$\text{let } y = x^m$$

$$\{m(m-1) - m + 1\} x^m = 0$$

$$m^2 - 2m + 1 = 0$$

$$m = 1$$

$$\therefore y = c_1 x + c_2 x \ln x \quad \leftarrow y(1) = 1.5$$

$$1.5 = c_1$$

$$y' = c_1 + c_2 \ln x + c_2 \quad \leftarrow y'(1) = 0.25$$

$$0.25 = c_1 + c_2$$

$$\therefore c_2 = -1.25$$

$$\therefore y = 1.5x - 1.25x \ln x$$

2.6

2.  $e^{4x}, e^{-1.5x}$

$$W = \begin{vmatrix} e^{4x} & e^{-1.5x} \\ 4e^{4x} & -1.5e^{-1.5x} \end{vmatrix} = -1.5e^{2.5x} - 4e^{2.5x} \\ = -5.5e^{2.5x} \neq 0 \\ \therefore \text{linearly Independent}$$

5.  $x^3, x^2$

$$W = \begin{vmatrix} x^3 & x^2 \\ 3x^2 & 2x \end{vmatrix} = 2x^4 - 3x^4 \\ = -x^4 \neq 0 \\ \therefore \text{linearly Independent}$$

9.  $\cos 5x, \sin 5x, y(0)=3, y'(0)=-5$

a) If  $y = e^{5ix}$   
 $y = \cos 5x + i \sin 5x$

$$\therefore \lambda = \pm 5i$$

$$\lambda^2 = -25$$

$$\lambda^2 + 25 = 0$$

$$y'' + 25y = 0$$

b)

$$W = \begin{vmatrix} \cos 5x & \sin 5x \\ -5\sin 5x & 5\cos 5x \end{vmatrix} = 5(\cos^2 5x + \sin^2 5x) \\ = 5 \neq 0$$

$$\therefore \text{linearly Independent}$$

c)  $y = c_1 \cos 5x + c_2 \sin 5x \leftarrow y(0)=3$

$$3 = c_1$$

$$y' = -5c_1 \sin 5x + 5c_2 \cos 5x \leftarrow y'(0) = -5$$

$$-5 = 5c_2$$

$$-1 = c_2$$

$$\therefore y = 3\cos 5x - \sin 5x$$

2.7

1.  $y'' + 5y' + 6y = 2e^x$

to get  $y_h$ , let  $y = e^{\lambda x}$ 

$$(\lambda^2 + 5\lambda + 6)e^{\lambda x} = 0$$

$$\lambda = -2, -3$$

$$\therefore y_h = c_1 e^{-2x} + c_2 e^{-3x}$$

to get  $y_p$  let  $y = Ae^x$ 

$$Ae^x - 5Ae^x + 6Ae^x = 2e^x$$

$$\therefore A = 1$$

$$\therefore y_p = e^x$$

$$\therefore y = y_h + y_p = c_1 e^{-2x} + c_2 e^{-3x} + e^x$$

3.  $y'' + 3y' + 2y = 12x^2$

to get  $y_h$ , let  $y = e^{\lambda x}$ 

$$(\lambda^2 + 3\lambda + 2)e^{\lambda x} = 0$$

$$\lambda = -1, -2$$

$$\therefore y_h = c_1 e^{-x} + c_2 e^{-2x}$$

to get  $y_p$ , let  $y = k_2 x^2 + k_1 x + k_0$ 

$$y' = 2k_2 x + k_1$$

$$y'' = 2k_2$$

$$\therefore 2k_2 + 3(2k_2 x + k_1) + 2(k_2 x^2 + k_1 x + k_0) = 12x^2$$

$$2k_2 x^2 + (6k_2 + 2k_1)x + 2k_2 + 3k_1 + 2k_0 = 12x^2$$

$$k_2 = 6$$

$$k_1 = 18$$

$$k_0 = -21$$

$$\therefore y_p = 6x^2 + 18x - 21$$

$$\therefore y = y_h + y_p = c_1 e^{-x} + c_2 e^{-2x} + 6x^2 + 18x - 21$$

5.  $y'' + 4y' + 4y = e^{2x} \cos x$

to get  $y_h$ , let  $y = e^{\lambda x}$ 

$$(\lambda^2 + 4\lambda + 4)e^{\lambda x} = 0$$

$$\lambda = -2$$

$$\therefore y_h = c_1 e^{-2x} + c_2 x e^{-2x} \leftarrow \text{Reduction of order} \uparrow$$

$$y_h = c_1 e^{-2x} + c_2 x e^{-2x}$$

let  $y = e^{2x}(A \cos x + B \sin x)$  for  $y_p$

$$y' = -e^{2x}(A \cos x + B \sin x) + e^{2x}(-A \sin x + B \cos x)$$

$$y'' = e^{2x}(A \cos x + B \sin x) - e^{2x}(-A \sin x + B \cos x) - e^{2x}(-A \sin x + B \cos x) + e^{2x}(-A \cos x - B \sin x)$$

$$= 2e^{2x}(-A \sin x + B \cos x) - 4(A \cos x + B \sin x) + 4(-A \sin x + B \cos x) + 4(A \cos x + B \sin x) = 2e^{2x}(-A \sin x + B \cos x)$$

$$\therefore -2B \cos x + 4A \cos x + 4B \sin x + 4A \cos x = \cos x$$

$$8A + 2B = 1$$

$$2A - 4B - 4A + 4B = 0$$

$$A = 0$$

$$B = \frac{1}{2}$$

$$\therefore y_p = \frac{1}{2} e^{2x} \cos x$$

$$\therefore y = y_h + y_p = c_1 e^{-2x} + c_2 x e^{-2x} + \frac{1}{2} e^{2x} \cos x$$

11.  $y'' + 4y = 8x^2$ ,  $y(0) = -3$ ,  $y'(0) = 0$

let  $y = e^{\lambda x}$  for  $y_h$ 

$$e^{\lambda x}(\lambda^2 + 4) = 0$$

$$\lambda = \pm 2i$$

$$\therefore y_h = c_1 \cos 2x + c_2 \sin 2x$$

let  $y = k_2 x^2 + k_1 x + k_0$  for  $y_p$ 

$$y' = 2k_2 x + k_1$$

$$y'' = 2k_2$$

$$\therefore 2k_2 + 4(k_2 x^2 + k_1 x + k_0) = 8x^2$$

$$4k_2 x^2 + 4k_1 x + 2k_2 + 4k_0 = 8x^2$$

$$k_2 = 2$$

$$k_1 = 0$$

$$k_0 = -4$$

$$\therefore y_p = 2x^2 - 4$$

$$\therefore y = y_h + y_p = c_1 \cos 2x + c_2 \sin 2x + 2x^2 - 4 \quad (y(0) = -3)$$

$$-3 = c_1 - 4$$

$$1 = c_1$$

$$y' = -2c_1 \sin 2x + 2c_2 \cos 2x + 4x \quad (y'(0) = 0)$$

$$\therefore c_2 = 0$$

$$\therefore y = \cos 2x + 2x^2 - 4$$

$$14. y'' + 6y' + 9y = e^{-x} \cos 2x$$

$$\text{Let } y = e^{\lambda x} \text{ for } y_h$$

$$(\lambda^2 + 6\lambda + 9)e^{\lambda x} = 0$$

$$\lambda = -3$$

$$\therefore y_h = C_1 e^{-3x} + C_2 x e^{-3x}$$

$$\text{Let } y = e^{-x} (A \cos 2x + B \sin 2x)$$

$$y' = -e^{-x} (A \cos 2x + B \sin 2x) + 2e^{-x} (-A \sin 2x + B \cos 2x)$$

$$y'' = e^{-x} (A \cos 2x + B \sin 2x) - 2e^{-x} (-A \sin 2x + B \cos 2x) - 2e^{-x} (-A \sin 2x + B \cos 2x) + 4e^{-x} (-A \cos 2x - B \sin 2x)$$

$$= e^{-x} (A \cos 2x + B \sin 2x) - 4e^{-x} (-A \sin 2x + B \cos 2x) + 4e^{-x} (-A \cos 2x - B \sin 2x)$$

$$e^{-x} (A \cos 2x + B \sin 2x) - 4e^{-x} (-A \sin 2x + B \cos 2x) + 4e^{-x} (-A \cos 2x - B \sin 2x)$$

$$+ 6 \{ -e^{-x} (A \cos 2x + B \sin 2x) + 2e^{-x} (-A \sin 2x + B \cos 2x) \}$$

$$+ 9 e^{-x} (A \cos 2x + B \sin 2x)$$

$$= e^{-x} \cos 2x$$

$$A \cos \quad B \sin$$

$$\therefore y_p = \frac{1}{2} e^{-x} \sin 2x$$

$$-4B \cos \quad 4A \sin$$

$$y = y_h + y_p = C_1 e^{-3x} + C_2 x e^{-3x} + \frac{1}{2} e^{-x} \sin 2x \quad (y(0) = 1)$$

$$-4A \cos \quad -4B \sin$$

$$C_1 = 1$$

$$-6A \cos \quad -6B \sin$$

$$y' = -3C_1 e^{-3x} + C_2 e^{-3x} + C_2 x (-3) e^{-3x} - \frac{1}{2} e^{-x} \sin 2x + e^{-x} \cos 2x \quad (y'(0) = -1)$$

$$2B \cos \quad -2A \sin$$

$$-3C_1 + C_2 + 1 = -1$$

$$9A \cos \quad 9B \sin$$

$$C_2 = 1$$

$$\cos$$

$$\therefore y = e^{-3x} + x e^{-3x} + \frac{1}{2} e^{-x} \sin 2x$$

$$-2B = 1$$

$$B = -\frac{1}{2} \quad A = 0$$

```

function xdot = sdof(t,x);

m = 1; k = 5; c = 6; w = 50;

xdot = zeros(2,1);

xdot(1) = x(2);

xdot(2) = -(k/m)*x(1)-(c/m)*x(2)+40*cos(w*t);

end

```

```

[t,x] = ode45(@sdof,[0 10],[0 1]);

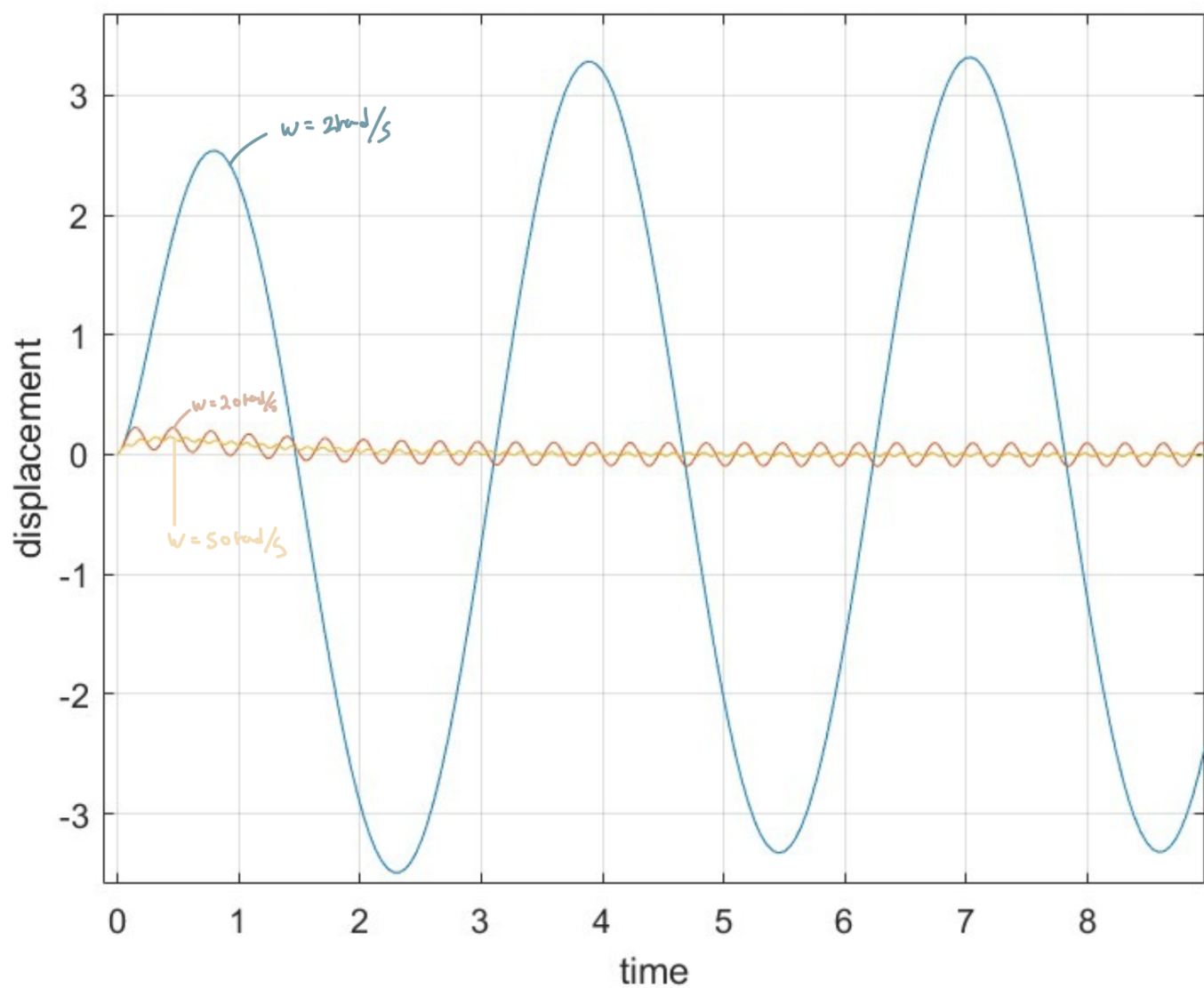
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```

plot(t,x(:,1));

```

를 작성하고  $w$ 를 바꾸며 값을 얻었다



2.10

$$2. y'' + 9y = \frac{1}{\sin 3x}$$

$$\text{let } y_h = e^{\lambda x}$$

$$\lambda^2 + 9 = 0$$

$$\lambda = \pm 3i$$

$$\therefore y_h = C_1 \cos 3x + C_2 \sin 3x$$

$$W = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} = 3$$

$$W_1 = \begin{vmatrix} 0 & \sin 3x \\ 1 & 3\cos 3x \end{vmatrix} = -\sin 3x$$

$$W_2 = \begin{vmatrix} \cos 3x & 0 \\ -3\sin 3x & 1 \end{vmatrix} = \cos 3x$$

$$y_p = \cos 3x \int \frac{-\sin 3x}{3} x \frac{1}{\sin 3x} dx + \sin 3x \int \frac{\cos 3x}{3} x \frac{1}{\sin 3x} dx$$

$$= \frac{1}{3} x \cos 3x + \frac{1}{9} \sin 3x \ln |\sin 3x|$$

$$\therefore y = C_1 \cos 3x + C_2 \sin 3x + \frac{1}{3} x \cos 3x + \frac{1}{9} \sin 3x \ln |\sin 3x|$$

$$3. x^2 y'' - xy' - 3y = x^2 \rightarrow y'' - \frac{y'}{x} - \frac{3}{x^2} y = 1$$

$$\text{let } y_h = x^m$$

$$m(m-1) - m - 3 = 0$$

$$m^2 - 2m - 3 = 0$$

$$m = -1, 3$$

$$\therefore y_h = C_1 x^{-1} + C_2 x^3$$

$$W = \begin{vmatrix} x^{-1} & x^3 \\ -x^{-2} & 3x^2 \end{vmatrix} = 4x$$

$$\frac{W_1}{W} = -\frac{x^2}{4} \quad \frac{W_2}{W} = \frac{1}{4x^2}$$

$$W_1 = \begin{vmatrix} 0 & x^3 \\ 1 & 3x^2 \end{vmatrix} = -x^3$$

$$W_2 = \begin{vmatrix} x^{-1} & 0 \\ -x^{-2} & 1 \end{vmatrix} = x^{-1}$$

$$\therefore y_p = x^{-1} \int -\frac{x^2}{4} dx + x^3 \int \frac{1}{4x^2} dx$$

$$= -\frac{x^2}{3}$$

$$\therefore y = C_1 x^{-1} + C_2 x^3 - \frac{x^2}{3}$$

$$4. y'' - 4y' + 5y = e^{2x} \cos x$$

$$\text{let } y_h = e^{\lambda x}$$

$$\lambda^2 - 4\lambda + 5 = 0$$

$$\lambda = 2 \pm i$$

$$y_h = e^{2x} (C_1 \cos x + C_2 \sin x)$$

$$W = \begin{vmatrix} e^{2x} \cos x & e^{2x} \sin x \\ 2e^{2x} \cos x & 2e^{2x} \sin x \\ -e^{2x} \sin x & +e^{2x} \cos x \end{vmatrix} = e^{4x}$$

$$W_1 = \begin{vmatrix} 0 & e^{2x} \sin x \\ 2e^{2x} \sin x & +e^{2x} \cos x \end{vmatrix} = -e^{2x} \sin x$$

$$W_2 = \begin{vmatrix} e^{2x} \cos x & 0 \\ 2e^{2x} \cos x & 1 \\ -e^{2x} \sin x & 1 \end{vmatrix} = e^{2x} \cos x$$

$$y_p = e^{2x} \cos x \int \frac{-e^{2x} \sin x}{e^{4x}} x e^{2x} dx$$

$$+ e^{2x} \sin x \int \frac{e^{2x} \cos x}{e^{4x}} x e^{2x} dx$$

$$= -e^{2x} \cos x x^2 + e^{2x} \sin x x \ln |\sin x|$$

$$\therefore y = e^{2x} (C_1 \cos x + C_2 \sin x - x^2 \cos x + \sin x \ln |\sin x|)$$

↑

## 2 Review

7.  $16y'' + 56y' + 45y = 0$

let  $y = e^{\lambda x}$

$(16\lambda^2 + 56\lambda + 45)e^{\lambda x} = 0$

$\frac{-28 \pm \sqrt{28^2 - 16 \cdot 45}}{8} = \frac{-28 \pm 8}{8} = \frac{-14}{2}, \frac{5}{2}$

$\therefore y = C_1 e^{-\frac{14}{2}x} + C_2 e^{\frac{5}{2}x}$

9.  $y'' + 4y' + 13y = 0$

let  $y = e^{\lambda x}$

$(\lambda^2 + 4\lambda + 13)e^{\lambda x} = 0$

$\lambda = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$

$\therefore y = e^{-2x}(A \cos(3x) + B \sin(3x))$

11.  $(9D^2 - 12D + 4I)y = 0$

$9y'' - 12y' + 4y = 0$

let  $y = e^{\lambda x}$

$9\lambda^2 - 12\lambda + 4 = 0$

$(3\lambda - 2)^2 = 0$

$\lambda = \frac{2}{3}$

$\therefore y = C_1 e^{\frac{2}{3}x} + C_2 x e^{\frac{2}{3}x}$

13.  $(x^2 D^2 + 2xD - 12I)y = 0$

$x^2 y'' + 2xy' - 12y = 0$

let  $y = x^n$

$(n(n+1) + 2n - 12)x^n = 0$

$n^2 + n - 12 = 0$

$n = -4, 3$

$\therefore y = C_1 x^{-4} + C_2 x^3$

15.  $(2D^2 - 3D - 2I)y = 3 - 2x^2$

let  $y = e^{\lambda x}$  for  $y_h$

$(2\lambda^2 - 3\lambda - 2) = 0$

$(2\lambda + 1)(\lambda - 2) = 0$

$\therefore \lambda = -\frac{1}{2}, 2$

$\rightarrow y = C_1 e^{-\frac{1}{2}x} + C_2 e^{2x}$

let  $y = k_2 x^2 + k_1 x + k_0$  for  $y_p$

$y' = 2k_2 x + k_1$

$y'' = 2k_2$

$\therefore 4k_2 - 3(2k_2 x + k_1) - 2(k_2 x^2 + k_1 x + k_0) = 3 - 2x^2$

$-2k_2 x^2 - (6k_2 + 2k_1)x + 4k_2 - 3k_1 - 2k_0 = 3 - 2x^2$

$\therefore k_2 = 1$

$k_1 = -3$

$k_0 = \frac{13}{2}$

$\therefore y = C_1 e^{-\frac{1}{2}x} + C_2 e^{2x} + x^2 - 3x + \frac{13}{2}$

## Chapter 2.

6.  $1, x^2, x^4, x^2 y''' - 3xy'' + 3y' = 0$

$$y = 1 \Rightarrow y' = 0$$

$$y' = 0$$

$$y'' = 0$$

$$y''' = 0$$

$$y = x^2 \Rightarrow y' = 2x$$

$$y' = 2x$$

$$y'' = 2$$

$$y''' = 0$$

$$y = x^4$$

$$y' = 4x^3$$

$$y'' = 12x^2$$

$$y''' = 24x$$

$$\therefore x^2 y''' - 3xy'' + 3y' = 0$$

$$0 = 0$$

$$x^2 y''' - 3xy'' + 3y' = 0$$

$$-6x + 6x = 0$$

$$0 = 0$$

$$x^2 y''' - 3xy'' + 3y' = 0$$

$$24x^3 - 36x^3 + 12x^3 = 0$$

$$0 = 0$$

$$W = \begin{vmatrix} 1 & x^2 & x^4 \\ 0 & 2x & 4x^3 \\ 0 & 2 & 12x^2 \end{vmatrix} = 24x^3 - 8x^3 - 16x^3 \neq 0$$

$\therefore$  linearly Independent

$\therefore$  3개의 함수는 basis 이다.

10.  $e^{2x}, xe^{2x}, x^2 e^{2x}$

$$W = \begin{vmatrix} e^{2x} & xe^{2x} & x^2 e^{2x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} & 2xe^{2x} + 2x^2 e^{2x} \\ 4e^{2x} & 4e^{2x} + 4xe^{2x} & (2 + 8x + 4x^2)e^{2x} \end{vmatrix} = e^{6x} \begin{vmatrix} 1 & x & x^2 \\ 2 & 1+2x & 2x+2x^2 \\ 4 & 1+4x & 2+8x+4x^2 \end{vmatrix}$$

$$= (1+2x)(2+8x+4x^2) - (2x+2x^2)(4+4x)$$

$$- x \{ 2(2+8x+4x^2) - (2x+2x^2) \times 4 \}$$

$$+ x^2 (2(4+4x) - 4(1+2x))$$

$$x^3 \text{의 계수 } 0$$

$$x^2 \text{의 계수 } 0$$

$$x \text{의 계수 } 0$$

$$\text{상수} : 2$$

$$\therefore W \neq 0$$

$\therefore$  linearly independent

15.  $\cosh x, \sinh x, e^x$

$$W = \begin{vmatrix} \cosh x & \sinh x & e^x \\ \sinh x & \cosh x & e^x \\ \cosh x & \sinh x & e^x \end{vmatrix} = \cosh x (e^x \cosh x - e^x \sinh x) = e^x (C^2 - S^2 - S^2 + S^2) + e^x$$

$$- \sinh x (e^x \sinh x - e^x \cosh x) = 2e^x$$

$$+ e^x (\sinh^2 x - \cosh^2 x) \therefore W \neq 0$$

$\therefore$  linearly independent



3.2

$$1. \quad y''' + 9y' = 0$$

$$\text{let } y = e^{\lambda x}$$

$$(\lambda^3 + 9\lambda)e^{\lambda x} = 0$$

$$\lambda(\lambda^2 + 9) = 0$$

$$\lambda = 0, \pm 3i$$

$$\therefore y = c_1 \cos 3x + c_2 \sin 3x + c_3$$

$$4. \quad (D^3 - D^2 - D + I)y = 0$$

$$\text{let } y = e^{\lambda x}$$

$$\lambda^3 - \lambda^2 - \lambda + 1 = 0$$

$$\begin{array}{c} 1 \left( \begin{array}{ccc|c} 1 & -1 & -1 & 1 \\ & 1 & 0 & -1 \\ \hline 1 & 0 & -1 & 0 \\ & 1 & 1 & 1 \\ \hline 1 & 1 & 0 & 0 \end{array} \right) \end{array}$$

$$\therefore (\lambda + 1)(\lambda - 1)^2 = 0$$

$$\lambda = -1 \quad \text{and} \quad \lambda = 1 \quad (\text{repeated})$$

$$\therefore y = c_1 e^x + c_2 x e^x + c_3 e^{-x}$$

$$10. \quad y^{(4)} - 14y'' = 0 \quad y(0) = \frac{1}{2}, \quad y'(0) = -\frac{3}{2}, \quad y''(0) = \frac{5}{2}, \quad y'''(0) = -\frac{7}{2}$$

$$\text{let } y = e^{\lambda x}$$

$$\lambda^4 + 4 = 0$$

$$\lambda^2 = \pm 2i$$

$$\lambda = \pm i, \pm i$$

$$\therefore \text{basis: } e^x \cos x, e^x \sin x, e^{-x} \cos x, e^{-x} \sin x$$

$$\therefore y = c_1 e^x \cos x + c_2 e^x \sin x + c_3 e^{-x} \cos x + c_4 e^{-x} \sin x \quad y(0) = \frac{1}{2}$$

$$\frac{1}{2} = c_1 + c_3$$

$$y' = c_1 (e^x \cos x - e^x \sin x) + c_2 (e^x \sin x + e^x \cos x) + c_3 (-e^x \cos x - e^x \sin x) + c_4 (-e^x \sin x + e^x \cos x) \quad y'(0) = -\frac{3}{2}$$

$$-\frac{3}{2} = c_1 + c_2 - c_3 + c_4$$

$$y'' = c_1 (-2e^x \sin x) + c_2 (2e^x \cos x) + c_3 (2e^x \sin x) + c_4 (-2e^x \cos x) \quad y''(0) = \frac{5}{2}$$

$$\frac{5}{2} = 2c_2 - 2c_4$$

$$y''' = -2c_1 (e^x \sin x + e^x \cos x) + 2c_2 (e^x \cos x - e^x \sin x) + 2c_3 (-e^x \sin x + e^x \cos x) + 2c_4 (-e^x \cos x - e^x \sin x) \quad y'''(0) = -\frac{7}{2}$$

$$-\frac{7}{2} = -2c_1 + 2c_2 + 2c_3 - 2c_4$$

$$\frac{1}{2} = C_1 + C_3$$

$$-\frac{3}{2} = C_1 + C_2 - C_3 + C_4$$

$$\frac{5}{2} = 2C_2 - 2C_4$$

$$-\frac{7}{2} = -2C_1 + 2C_2 + 2C_3 - 2C_4$$

$$C_1 = \frac{5}{16}, C_2 = -\frac{3}{16}, C_3 = \frac{3}{16}, C_4 = -\frac{23}{16}$$

$$\therefore y = C_1 e^x \cos x + C_2 e^x \sin x + C_3 e^{-x} \cos x + C_4 e^{-x} \sin x$$

$$= \frac{5}{16} e^x \cos x - \frac{3}{16} e^x \sin x + \frac{3}{16} e^{-x} \cos x - \frac{23}{16} e^{-x} \sin x$$

33

$$1. y''' - 3y'' + 3y' - y = e^x - x - 1$$

$$2. y''' + 2y'' - y' - 2y = 1 - 4x^2$$

for  $y_h$ , let  $y = e^{\lambda x}$

$$\lambda^3 + 2\lambda^2 - \lambda - 2 = 0$$

$$(\lambda - 1)(\lambda + 1)(\lambda + 2) = 0$$

$$\lambda = -2, -1, 1$$

$$\therefore y_h = C_1 e^{-2x} + C_2 e^{-x} + C_3 e^x$$

for  $y_p$  let  $y = k_2 x^2 + k_1 x + k_0$

$$y' = 2k_2 x + k_1$$

$$y'' = 2k_2$$

$$y''' = 0 \quad \text{OK!}$$

$$4k_2 - 2k_2 x - k_1 - 2k_2 x^2 - 2k_1 x - 2k_0 = 1 - 4x^2$$

$$-2k_2 x^2 + (-2k_2 - 2k_1)x + 4k_2 - k_1 - 2k_0 = 1 - 4x^2$$

$$k_2 = 2$$

$$-2k_2 - 2k_1 = 0 \rightarrow k_1 = -2$$

$$4k_2 - k_1 - 2k_0 = 1$$

$$k_0 = -\frac{5}{2}$$

$$\therefore y_p = 2x^2 + x - \frac{5}{2}$$

$$\therefore y = C_1 e^{-2x} + C_2 e^{-x} + C_3 e^x + 2x^2 + x - \frac{5}{2}$$

$$3. (D^4 + 5D^2 + 4I)y = 3.5 \sin 2x$$

let  $y = e^{\lambda x}$  for  $y_h$

$$\lambda^4 + 5\lambda^2 + 4$$

$$\lambda^2 = -1, -4$$

$$\lambda = \pm i, \pm 2i$$

basis:  $\cos x, \sin x, \cos 2x, \sin 2x$

$$\therefore y_h = C_1 \cos x + C_2 \sin x + C_3 \cos 2x + C_4 \sin 2x$$

Next page

$$\text{let } y = A \sinh 2x \quad \text{for } y_p$$

$$y'' = 4A \sinh 2x$$

$$y^{(4)} = 16A \sinh 2x \quad \text{Dif}$$

$$16A \sinh 2x + 20A \sinh 2x + 4A \sinh 2x = 3.5 \sinh 2x$$

$$\therefore A = 0.6875$$

$$\therefore y_p = 0.6875 \sinh 2x$$

$$\therefore y = C_1 \cos x + C_2 \sin x + C_3 \cos 2x + C_4 \sin 2x + 0.6875 \sinh 2x$$

$$5. (x^3 D^3 + 2x^2 D^2 - xD + I)y = x^{-2}$$

$$\text{let } y = x^m \quad \text{for } y_h$$

$$m(m-1)(m-2) + 2m(m-1) - m + 1 = 0$$

$$m^3 - 3m^2 + 2m + 2m^2 - 2m - m + 1 = 0$$

$$m^3 - m^2 - m + 1 = 0$$

$$\begin{array}{c|ccc} 1 & 1 & -1 & -1 & 1 \\ & & 1 & 0 & -1 \\ \hline & 1 & 0 & -1 & 0 \end{array}$$

$$\therefore x = -1, 1, 1$$

$$\therefore y_h = C_1 x^{-1} + C_2 x + C_3 x \ln|x|$$

$$W = \begin{vmatrix} x^{-1} & x & x \ln|x| \\ -x^{-2} & 1 & \ln|x| + 1 \\ 2x^{-3} & 0 & \frac{1}{x} \end{vmatrix} = 4x^{-2} \quad W_1 = \begin{vmatrix} 0 & x & x \ln|x| \\ 0 & 1 & \ln|x| + 1 \\ 1 & 0 & \frac{1}{x} \end{vmatrix} = x(\ln|x| + 1) - x \ln|x| = x$$

$$W_2 = \begin{vmatrix} x^{-1} & 0 & x \ln|x| \\ -x^{-2} & 0 & \ln|x| + 1 \\ 2x^{-3} & 1 & \frac{1}{x} \end{vmatrix} = -\{x^{-1}(\ln|x| + 1) + x^{-2}(x \ln|x|)\} \quad W_3 = \begin{vmatrix} x^{-1} & x & 0 \\ -x^{-2} & 1 & 0 \\ 2x^{-3} & 0 & 1 \end{vmatrix} = x^{-1} + x^{-1} = 2x^{-1}$$

$$\therefore y_p = x^{-1} \int \frac{x^{-4}}{4x^{-2}} dx + x \int \frac{-2x^{-4} \ln|x| - x^{-6}}{4x^{-2}} dx + x \ln|x| \int \frac{2x^{-6}}{4x^{-2}} dx$$

$$= -\frac{x^{-2}}{4} + \frac{1}{6} x^{-2} \ln|x| + \frac{1}{18} x^{-2} + \frac{1}{12} x^{-2} - \frac{1}{6} x^{-2} \ln|x|$$

$$= -\frac{1}{9} x^{-2}$$

$$\therefore y = C_1 x^{-1} + C_2 x + C_3 x \ln|x| - \frac{1}{9} x^{-2}$$

$$8. y^{(4)} - 5y'' + 4y = 10e^{-3x}, \quad y(0) = 1, \quad y'(0) = 0, \quad y''(0) = 0, \quad y'''(0) = 0$$

$$\text{let } y = e^{\lambda x}$$

$$\lambda^4 - 5\lambda^2 + 4 = 0$$

$$\lambda = \pm 1, \pm 2 \quad \therefore y_h = C_1 e^{-x} + C_2 e^{-2x} + C_3 e^x + C_4 e^{2x}$$

Cliff

let  $y = Ae^{-3x}$  for  $y_p$

$$y' = (-3)Ae^{-3x}$$

$$y'' = (-3)^2 Ae^{-3x}$$

$$y''' = (-3)^3 Ae^{-3x}$$

$$y^{(4)} = (-3)^4 Ae^{-3x}$$

plug in  $A = \frac{1}{4}$

$$\therefore y_p = \frac{1}{4}e^{-3x}$$

$$\therefore y = c_1 e^{-2x} + c_2 e^{-x} + c_3 e^x + c_4 e^{2x} + \frac{1}{4}e^{-3x} \quad y(0) = 1$$

$$c_1 + c_2 + c_3 + c_4 + \frac{1}{4} = 1$$

$$y' = -2c_1 e^{-2x} - c_2 e^{-x} + c_3 e^x + 2c_4 e^{2x} - \frac{3}{4}e^{-3x} \quad y'(0) = 0$$

$$-2c_1 - c_2 + c_3 + 2c_4 - \frac{3}{4} = 0$$

$$y'' = 4c_1 e^{-2x} + c_2 e^{-x} + c_3 e^x + 4c_4 e^{2x} + \frac{9}{4}e^{-3x} \quad y''(0) = 0$$

$$4c_1 + c_2 + c_3 + 4c_4 + \frac{9}{4} = 0$$

$$y''' = -8c_1 e^{-2x} - c_2 e^{-x} + c_3 e^x + 8c_4 e^{2x} - \frac{27}{4}e^{-3x} \quad y'''(0) = 0$$

$$-8c_1 - c_2 + c_3 + 8c_4 - \frac{27}{4} = 0$$

plug in

$$c_1 = -\frac{5}{6}, c_2 = \frac{4}{3}, c_3 = \frac{1}{12}, c_4 = \frac{1}{6}$$

$$\therefore y = -\frac{5}{6}e^{-2x} + \frac{4}{3}e^{-x} + \frac{1}{12}e^x + \frac{1}{6}e^{2x} + \frac{1}{4}e^{-3x} = 0$$

13  $(D^3 - 4D)y = 10 \cos x + 5 \sin x \quad y(0) = 3, y'(0) = -2, y''(0) = -1$

let  $y = e^{\lambda x}$

$$\lambda^3 - 4\lambda = 0$$

$$\lambda = -2, 0, 2$$

$$\therefore y_h = c_1 e^{-2x} + c_2 e^{2x} + c_3$$

for  $y_p$ , let  $y = A \cos x + B \sin x$

$$y' = -A \sin x + B \cos x$$

$$y'' = A \cos x - B \sin x$$

$$A \cos x - B \sin x + 4A \sin x - 4B \cos x = 10 \cos x + 5 \sin x$$

$$A = 1, B = -2$$

$$\therefore y_p = (\cos x - 2 \sin x)$$

$$y = c_1 e^{-2x} + c_2 e^{2x} + c_3 + (\cos x - 2 \sin x) \quad y(0) = 3$$

$$c_1 + c_2 + c_3 = 3$$

$$y' = -2c_1 e^{-2x} + 2c_2 e^{2x} - \sin x - 2 \cos x \quad y'(0) = -2$$

$$= -2c_1 + 2c_2 - 2 = -2$$

$$y'' = 4c_1 e^{-2x} + 4c_2 e^{2x} - \cos x + 2 \sin x \quad y''(0) = -1$$

$$4c_1 + 4c_2 - 1 = -1$$

plug in

$$c_1 = 0, c_2 = 0, c_3 = 2$$

$$\therefore y = 2 + \cos x - 2 \sin x$$



# Review

$$6. y^{(4)} - 3y'' - 4y = 0$$

$$\text{let } y = e^{\lambda x}$$

$$\lambda^4 - 3\lambda^2 - 4 = 0$$

$$\lambda^2 = 4, \lambda^2 = -1$$

$$\lambda = \pm 2, \pm i$$

$$\therefore y = C_1 \cos x + C_2 \sin x + C_3 e^{2x} + C_4 e^{-2x}$$

$$10. x^2 y''' + 3x y'' - 2y' = 0$$

$$\text{let } y = x^m$$

$$m(m-1)(m-2) + 3m(m-1) - 2m = 0$$

$$m^3 - 3m^2 + 2m + 3m^2 - 3m - 2m = 0$$

$$m^3 - 3m = 0$$

$$m(m^2 - 3) = 0$$

$$m = -\sqrt{3}, 0, \sqrt{3}$$

$$\therefore y = C_1 x^{-\sqrt{3}} + C_2 x^{\sqrt{3}} + C_3$$

$$20. (D^3 + 3D^2 + 3D + 1)y = 8 \sin x, y(0) = -1, y'(0) = -3, y''(0) = 5$$

$$\text{let } y = e^{\lambda x} \text{ for } y_h$$

$$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$$

$$\lambda = -1, -1, -1$$

$$\therefore y_h = C_1 e^{-x} + C_2 x e^{-x} + C_3 x^2 e^{-x}$$

$$\text{let } y = A \cos x + B \sin x \text{ for } y_p$$

$$y' = -A \sin x + B \cos x$$

$$y'' = -A \cos x - B \sin x$$

$$y''' = A \sin x - B \cos x$$

$$\text{Plug into equation}$$

$$A = -2, B = 2$$

$$\therefore y = C_1 e^{-x} + C_2 x e^{-x} + C_3 x^2 e^{-x} - 2 \cos x + 2 \sin x \quad y(0) = -1$$

$$C_1 - 2 = -1$$

$$C_2 - C_1 - 2 = -3$$

$$2C_3 - 2C_2 + C_1 + 2 = 5$$

$$y'(0) = -3$$

$$y''(0) = 5$$

$$\therefore C_1 = 1, C_2 = 0, C_3 = 1$$

$$\therefore y = e^{-x} + x^2 e^{-x} - 2 \cos x + 2 \sin x$$

4.

$$4.10 \quad y'' + 4y' + 3y = 0$$

$$y = y_1, \quad \dot{y} = y_2 = \dot{y}_1$$

$$a) \begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = -3y_1 - 4y_2 \end{cases}$$

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\dot{y} = Ay$$

$$\text{let } y = xe^{xt}$$

$$\text{then } \dot{y} = Ay$$

$$\lambda x e^{xt} = A x e^{xt}$$

$$\lambda x = Ax$$

$$(\lambda I - A)x = \vec{0}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ 3 & \lambda + 4 \end{vmatrix} = \lambda^2 + 4\lambda + 3 = 0$$

$$\therefore \lambda = -1, -3$$

$$x = -3$$

$$\therefore \begin{bmatrix} -1 & -1 \\ 3 & 3 \end{bmatrix} x_1 = 0$$

$$\begin{bmatrix} -3 & -1 \\ 3 & 1 \end{bmatrix} x_2 = 0$$

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{-3t}$$

$$12. \quad y''' - 2y'' - y' + 2y = 0$$

$$y = y_1$$

$$\dot{y} = \dot{y}_1 = y_2 \quad \text{or } 2y_2 \text{ and } 2y_3$$

$$\ddot{y} = \ddot{y}_1 = \dot{y}_2 = y_3$$

$$\dot{y}_1 = y_2$$

$$\dot{y}_2 = y_3$$

$$\dot{y}_3 = 2y_3 + y_2 - 2y_1$$

$$\dot{y} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{bmatrix} y$$

↑

$$\text{let } y = xe^{xt}$$

$$\dot{y} = Ay$$

$$\lambda x e^{xt} = A x e^{xt}$$

$$\lambda x = Ax$$

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 1 & 0 & 0 \\ 0 & \lambda & -1 \\ 2 & -1 & \lambda - 2 \end{vmatrix} = 0$$

$$\lambda(\lambda(\lambda-2)-1)+2=0$$

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$$

$$\lambda = -1, 1, 2$$

$$\therefore \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 2 & -1 & -3 \end{bmatrix} x_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \leftarrow \lambda = -1$$

$$x_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix} x_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \leftarrow \lambda = 1$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & -1 \\ 2 & -1 & 0 \end{bmatrix} x_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \leftarrow \lambda = 2$$

$$x_3 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2t}$$

$$4.3.1 \begin{cases} \dot{y}_1 = 2y_1 - y_2 \\ \dot{y}_2 = 3y_1 - 2y_2 \end{cases}$$

$$\dot{y} = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} y$$

$$\text{let } y = x e^{\lambda t}$$

$$\lambda x e^{\lambda t} = A x e^{\lambda t}$$

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 2 & 1 \\ -3 & \lambda + 2 \end{vmatrix} = 0$$

$$\lambda^2 - 4 + 3 = 0$$

$$\lambda = \pm 1$$

$$\lambda = -1, (\lambda I - A)x = \vec{0}$$

$$\begin{bmatrix} -3 & 1 \\ -3 & 1 \end{bmatrix} x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\lambda = 1, (\lambda I - A)x = \vec{0}$$

$$\begin{bmatrix} -1 & 1 \\ -3 & 3 \end{bmatrix} x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$$

$$6. \begin{cases} \dot{y}_1 = 2y_1 - 2y_2 \\ \dot{y}_2 = 2y_1 + 2y_2 \end{cases}$$

$$\dot{y} = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix} y$$

$$\text{let } y = x e^{\lambda t}$$

$$\lambda x e^{\lambda t} = A x e^{\lambda t}$$

$$\det(\lambda I - A) = 0$$

$$\begin{bmatrix} \lambda - 2 & 2 \\ -2 & \lambda - 2 \end{bmatrix} = \vec{0}$$

$$(\lambda - 2)^2 + 4 = 0$$

$$\lambda^2 - 4\lambda + 8$$

$$\lambda = 2 \pm 2i$$

$$\lambda = 2 - 2i$$

$$\begin{bmatrix} -2i & 2 \\ -2 & -2i \end{bmatrix} x_1 = \vec{0}$$

$$x_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\lambda = 2 + 2i$$

$$\begin{bmatrix} 2i & 2 \\ -2 & 2i \end{bmatrix} x_2 = \vec{0}$$

$$x_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{(2-2i)t} + c_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{(2+2i)t}$$

$$= c_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{2t} (\cos 2t - i \sin 2t)$$

$$+ c_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{2t} (\cos 2t + i \sin 2t)$$

$$8. \begin{cases} \dot{y}_1 = 8y_1 - y_2 \\ \dot{y}_2 = y_1 + 10y_2 \end{cases}$$

$$\dot{y} = \begin{bmatrix} 8 & -1 \\ 1 & 10 \end{bmatrix} y$$

$$\text{let } y = x e^{\lambda t}$$

$$\lambda x e^{\lambda t} = A x e^{\lambda t}$$

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 8 & 1 \\ -1 & \lambda - 10 \end{vmatrix} = 0$$

$$\lambda^2 - 18\lambda + 81 = 0$$

$$\lambda = 9 \text{ (32)}$$

$$\lambda = 9 \text{ only}$$

$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} x_1 = \vec{0}$$

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore y_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{9t}$$

$$\text{let } y_2 = x_1 e^{\lambda_1 t} + v e^{\lambda_2 t}$$

Not possible



$$\text{then } \dot{y}_2 = x_1 e^{\lambda_1 t} + x_1 t \lambda_1 e^{\lambda_1 t} + \lambda_1 y_1 e^{\lambda_1 t} = A x_1 t e^{\lambda_1 t} + A y_1 e^{\lambda_1 t}$$

$$(\lambda_1 I - A)V = -x_1$$

$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} V = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\therefore y_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} t e^{at} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{at}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{at} + c_2 \left( \begin{bmatrix} 1 \\ -1 \end{bmatrix} t e^{at} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{at} \right)$$

$$10. \begin{cases} \dot{y}_1 = -4y_1 + 5y_2 \\ \dot{y}_2 = -y_1 + 2y_2 \end{cases} \quad \begin{matrix} y_1(0) = 0 \\ y_2(0) = 4 \end{matrix}$$

$$\dot{y} = \begin{bmatrix} -4 & 5 \\ 1 & 2 \end{bmatrix} y$$

$$\text{let } y = x e^{\lambda t}$$

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda + 4 & -5 \\ 1 & \lambda - 2 \end{vmatrix} = 0$$

$$\lambda^2 + 2\lambda - 3 = 0$$

$$\lambda = -3, 1$$

$$\therefore (\lambda I - A)x = 0$$

$$\lambda = -3$$

$$\begin{bmatrix} 1 & -5 \\ 1 & -5 \end{bmatrix} x_1 = \vec{0}$$

$$x_1 = \begin{bmatrix} 1 \\ \frac{1}{5} \end{bmatrix}$$

$$\lambda = 1$$

$$\begin{bmatrix} 5 & -5 \\ 1 & -1 \end{bmatrix} x_2 = \vec{0}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ \frac{1}{5} \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$$

$$y_1 = c_1 e^{3t} + c_2 e^t \quad \leftarrow y_1(0) = 0$$

$$y_2 = c_1 \frac{1}{5} e^{3t} + c_2 e^t \quad \leftarrow y_2(0) = 4$$

$$c_1 + c_2 = 0$$

$$\frac{1}{5}c_1 + c_2 = 4$$

$$\therefore c_1 = -5, c_2 = 5$$

$$\therefore y = \begin{bmatrix} -5 \\ 1 \end{bmatrix} e^{3t} + \begin{bmatrix} 5 \\ 5 \end{bmatrix} e^t$$

4.4

$$2. \quad y_1' = -4y_1$$

$$y_2' = -3y_2$$

$$\dot{y} = \begin{bmatrix} -4 & 0 \\ 0 & -3 \end{bmatrix} y$$

$$\text{let } y = X e^{\lambda t}$$

$$\lambda X e^{\lambda t} = A X e^{\lambda t}$$

$$(\lambda I - A) X e^{\lambda t} = \vec{0}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda + 4 & 0 \\ 0 & \lambda + 3 \end{vmatrix} = (\lambda + 4)(\lambda + 3) = 0$$

$$\lambda = -4, -3$$

$$\lambda = -4$$

$$\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} X e^{\lambda t} = \vec{0}$$

$$X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = -3$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} X e^{\lambda t} = \vec{0}$$

$$X = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-3t}$$

$$5. \quad \dot{y}_1 = -y_1 + y_2$$

$$\dot{y}_2 = -y_1 - y_2$$

$$\dot{y} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} y$$

$$\text{let } y = X e^{\lambda t}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda + 1 & -1 \\ 1 & \lambda + 1 \end{vmatrix} = (\lambda + 1)^2 + 1 = 0$$

$$\lambda = -1 \pm i$$

$$\lambda = -1 \pm i$$

$$\lambda = -1 - i$$

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} X = \vec{0}$$

$$X = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\lambda = -1 + i$$

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} X = \vec{0}$$

$$X = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{-t} \cdot e^{-it} + c_2 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{-t} \cdot e^{it}$$

$$= c_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{-t} (\cos x - i \sin x) + c_2 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{-t} (\cos x + i \sin x)$$

$$8. \quad \dot{y}_1 = -y_1 + 4y_2$$

$$\dot{y}_2 = 3y_1 - 2y_2$$

$$\dot{y} = \begin{bmatrix} -1 & 4 \\ 3 & -2 \end{bmatrix} y$$

$$|\lambda I - A| = \begin{vmatrix} \lambda + 1 & -4 \\ -3 & \lambda + 2 \end{vmatrix} = 0$$

$$\lambda^2 + 3\lambda + 2 - 12 = 0$$

$$\lambda = -5, 2$$

$$\lambda = -5$$

$$\begin{bmatrix} -4 & -4 \\ -3 & -3 \end{bmatrix} X = \vec{0}$$

$$X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} 3 & -4 \\ -3 & 4 \end{bmatrix} X = \vec{0}$$

$$X = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} e^{2t}$$

$$11. \quad y'' + 2y' + 5y = 0$$

$$y = y_1$$

$$\dot{y}_1 = y_2$$

$$\dot{y}_1 = y_2$$

$$\dot{y}_2 = -5y_1 - 2y_2$$

$$\dot{y} = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix} y$$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ 5 & \lambda + 2 \end{vmatrix} = 0$$

$$\lambda(\lambda + 2) + 5 = 0$$

$$\lambda^2 + 2\lambda + 5 = 0$$

$$\lambda = -1 \pm 2i$$

$$\lambda = -1 - 2i$$

$$\begin{bmatrix} -1 - 2i & -1 \\ 5 & 1 - 2i \end{bmatrix} x = \vec{0}$$

$$x = \begin{bmatrix} 1 \\ 1 + 2i \end{bmatrix}$$

$$\lambda = -1 + 2i$$

$$\begin{bmatrix} -1 + 2i & -1 \\ 5 & 1 + 2i \end{bmatrix} x = \vec{0}$$

$$x = \begin{bmatrix} 1 \\ 1 - 2i \end{bmatrix}$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ 1 + 2i \end{bmatrix} e^{(-1-2i)t} + c_2 \begin{bmatrix} 1 \\ 1 - 2i \end{bmatrix} e^{(-1+2i)t}$$

$$y_1 = e^{-t} (c_1 \cos 2t + c_2 \sin 2t)$$

$$y_2 = e^{-t} (c_1 (1 + 2i) \cos 2t + c_2 (1 - 2i) \sin 2t)$$

4.6

$$S. \quad \dot{y}_1 = 4y_1 + 3y_2 + t$$

$$\dot{y}_2 = -2y_1 - y_2 - 2t$$

$$\dot{y} = \begin{bmatrix} 4 & 3 \\ -2 & -1 \end{bmatrix} y + \begin{bmatrix} t \\ -2t \end{bmatrix}$$

$$\text{let } y_h = x e^{\lambda t}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 4 & -3 \\ 2 & \lambda + 1 \end{vmatrix} = \lambda^2 - 3\lambda - 4 + 6$$

$$\lambda = 1, 2$$

$$\lambda = 1$$

$$\begin{bmatrix} -3 & -3 \\ 2 & 2 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} -2 & -3 \\ 2 & 3 \end{bmatrix}$$

$$x = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$\therefore y_h = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 3 \\ -2 \end{bmatrix} e^{2t}$$

$$\text{let } y_p = u + Vt$$

$$\dot{y} = V$$

$$\therefore V = \begin{bmatrix} 4 & 3 \\ -2 & -1 \end{bmatrix} (u + Vt) + \begin{bmatrix} t \\ -2t \end{bmatrix}$$

$$V_1 = 4u_1 + 3u_2 + 4V_1t + 3V_2t + t$$

$$\rightarrow 4V_1 + 3V_2 + 1 = 0, \quad 4u_1 + 3u_2 - V_1 = 0$$

$$V_2 = -2u_1 - u_2 - 2V_1t - V_2t - 2t$$

$$\rightarrow 2V_1 + V_2 + 2 = 0, \quad 2u_1 + u_2 + V_2 = 0$$

$$V_1 = \frac{5}{2}, V_2 = -3, u_1 = -\frac{13}{4}, u_2 = \frac{11}{2}$$

$$\therefore y_p = \begin{bmatrix} -\frac{13}{4} \\ \frac{11}{2} \end{bmatrix} + \begin{bmatrix} \frac{5}{2} \\ -3 \end{bmatrix} t$$

$$\therefore y = y_p + y_h = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 3 \\ -2 \end{bmatrix} e^{2t} + \dots$$

$$\dots + \begin{bmatrix} -\frac{13}{4} \\ \frac{11}{2} \end{bmatrix} + \begin{bmatrix} \frac{5}{2} \\ -3 \end{bmatrix} t$$

$$11. \dot{y}_1 = y_2 + 2e^t \quad y_1(0)=0, \quad y_2(0)=0$$

$$\dot{y}_2 = y_1 - 2e^t$$

$$\dot{y} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} y + \begin{bmatrix} 2 \\ -2 \end{bmatrix} e^t$$

$$\text{let } y_h = x e^{\lambda t}$$

$$(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - 1 = 0$$

$$\lambda = \pm 1$$

$$\lambda = -1$$

$$\lambda = 1$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} x = \vec{0} \quad \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} x = \vec{0}$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore y_h = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$$

$$\text{let } y_p = u t e^t + v e^t$$

$$\dot{y} = u e^t + u t e^t + v e^t$$

$$= (u t + v + u t) e^t$$

$$(u t + v + u t) e^t = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (u t + v) e^t + \begin{bmatrix} 2 \\ -2 \end{bmatrix} e^t$$

$$u t + v + u t = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (u t + v) + \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$u_1 t + v_1 + u_1 t = u_2 t + v_2 + 2 \rightarrow$$

$$u_2 t + v_2 + u_2 t = u_1 t + v_1 - 2 \rightarrow$$

$$u_1 = u_2, \quad u_1 + v_1 = v_2 + 2$$

$$u_2 + v_2 = v_1 - 2$$

$$v_1 - v_2 = v_2 - v_1 + 4$$

$$2v_1 - 4 = 2v_2$$

$$v_1 - 2 = v_2$$

$$\therefore y_p = \begin{bmatrix} v_1 \\ v_1 - 2 \end{bmatrix} e^t$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + \begin{bmatrix} v_1 \\ v_1 - 2 \end{bmatrix} e^t$$

$$\vec{y} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + \begin{bmatrix} v_1 \\ v_1 - 2 \end{bmatrix} e^t$$

$$y_1 = c_1 e^{-t} + c_2 e^t + v_1 e^t \quad y_1(0) = 0$$

$$0 = c_1 + c_2 + v_1$$

$$y_2 = -c_1 e^{-t} + c_2 e^t + (v_1 - 2) e^t \quad y_2(0) = 0$$

$$0 = -c_1 + c_2 + v_1 - 2$$

$$2c_1 + 2 = 0$$

$$c_1 = -1$$

$$c_2 = 1 - v_1$$

$$13 \quad \dot{y} = \begin{bmatrix} -5 & 5 \\ -2 & 1 \end{bmatrix} y$$

$$|\lambda I - X| = 0$$

$$\begin{vmatrix} \lambda + 5 & -5 \\ 2 & \lambda - 1 \end{vmatrix} = \lambda^2 + 4\lambda + 5$$

$$\lambda = -2 \pm i$$

$$\lambda = -2 - i$$

$$\lambda = -2 + i$$

$$\begin{bmatrix} 3-i & -5 \\ 2 & -3-i \end{bmatrix} x = \vec{0}$$

$$\begin{bmatrix} 3+i & -5 \\ 2 & -3+i \end{bmatrix} x_2 = \vec{0}$$

$$x_1 = \begin{bmatrix} 5 \\ 3-i \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 5 \\ 3+i \end{bmatrix}$$

$$y = 5c_1 e^{(-2-i)t} + 5c_2 e^{(-2+i)t}$$

$$= e^{-2t} \{ 5c_1 (\cos t - i \sin t) + 5c_2 (\cos t + i \sin t) \}$$

$$= e^{-2t} \{ (5c_1 + 5c_2) \cos t + (-5c_1 + 5c_2) i \sin t \}$$

$$y_2 = e^{-2t} \{ c_1 (3-i) (\cos t - i \sin t) + c_2 (3+i) (\cos t + i \sin t) \}$$

$$= e^{-2t} \{ \{ c_1 (3-i) + c_2 (3+i) \} (\cos t + i \{ c_1 (i-3) + c_2 (3+i) \} \sin t \}$$

$$21. \quad \dot{y}_1 = 2y_2$$

$$\dot{y}_2 = 2y_1 + 16t^2$$

$$\dot{y} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} y + \begin{bmatrix} 0 \\ 16t^2 \end{bmatrix}$$

$$\text{let } y_h = x e^{\lambda t}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -2 \\ -2 & \lambda \end{vmatrix} = \lambda^2 - 4 = 0$$

$$\lambda = -2, 2$$

$$\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} x_1 = \vec{0}$$

$$\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} x_2 = \vec{0}$$

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore y_h = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t}$$

$$\text{let } y_p = a + bt + ct^2$$

$$\dot{y} = b + 2ct$$

$$\rightarrow b + 2ct = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} (a + bt + ct^2) + \begin{bmatrix} 0 \\ 16t^2 \end{bmatrix}$$

$$b_1 + 2c_1 t = 2a_2 + 2b_2 t + 2c_2 t^2 \rightarrow c_2 = 0, c_1 = b_2, b_1 = 2b_2, a_2 = 0$$

$$b_2 + 2c_2 t = 2a_1 + 2b_1 t + 2c_1 t^2 + 16t^2$$

$$\rightarrow c_1 = -8, b_1 = c_2, b_2 = 2a_1, a_1 = -4$$

$$\therefore a_2 \begin{bmatrix} -4 \\ 0 \end{bmatrix}, b = \begin{bmatrix} 0 \\ -8 \end{bmatrix}, c = \begin{bmatrix} -8 \\ 0 \end{bmatrix}$$

$$\therefore y_p = \begin{bmatrix} -4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -8 \end{bmatrix} t + \begin{bmatrix} -8 \\ 0 \end{bmatrix} t^2$$

$$\therefore y = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + \begin{bmatrix} -4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -8 \end{bmatrix} t + \begin{bmatrix} -8 \\ 0 \end{bmatrix} t^2$$

$$22. \quad \dot{y}_1 = y_1 + y_2 + \sin t$$

$$\dot{y}_2 = 4y_1 + y_2$$

$$\dot{y} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} y + \begin{bmatrix} \sin t \\ 0 \end{bmatrix}$$

$$\text{let } y_h = x e^{\lambda t}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 1 & -1 \\ -4 & \lambda - 1 \end{vmatrix} = \lambda^2 - 2\lambda - 3 = 0$$

$$\lambda = -1, 3$$

$$\lambda = -1$$

$$\lambda = 3$$

$$\begin{bmatrix} -2 & -1 \\ -4 & -2 \end{bmatrix} x_1 = \vec{0}$$

$$\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix} x_2 = \vec{0}$$

$$x_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\therefore y_h = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t}$$

$$\text{let } y_p = a \cos t + b \sin t$$

$$\dot{y} = -a \sin t + b \cos t$$

$$\text{then } -a \sin t + b \cos t = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} (a \cos t + b \sin t) + \begin{bmatrix} \sin t \\ 0 \end{bmatrix}$$

